CarbonMindful: A Smart Carbon Footprint Calculator for AI Model Training

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Abstract

This research paper presents the design and assessment of CarbonMindful, a Smart Carbon Footprint Calculator for Training AI Models. CarbonMindful functions as a Streamlit application that calculates the environmental footprint associated with AI model training, thereby promoting responsible and environmentally friendly AI practices. Through real-time analysis of training telemetry, the system reconstructs energy consumption, emissions, and hardware utilization, enabling comparisons of regional scenarios and assumptions regarding data center efficiency.

In an evaluation involving a multilayer perceptron trained on the MNIST dataset over three epochs, the platform recorded energy consumption of 0.000625 kWh and emissions of 0.000269 kg CO_2e, achieving an accuracy of 95.14 percent. Although these values are derived from a lightweight benchmark workload and are not intended to represent large-scale training, they serve to demonstrate the platform's comparative and sensitivity analysis capabilities. These results should be interpreted as a methodological demonstration rather than as representative of industrial-scale AI training.

The selection of a lower-carbon grid resulted in an approximate 84 percent reduction in emissions, while an increase in PUE from 1.20 to 1.60 led to a marked rise in emissions. The metric of 0.000003 kg CO_2e per accuracy point underscores the diminishing returns associated with extended training durations. An analysis of hardware utilization revealed CPU saturation, suggesting that transitioning to more carbon-intensive hardware would yield minimal advantages. This AI platform demonstrates that environmental transparency and emissions-aware trade-offs can be seamlessly integrated into the routine practices of Machine Learning, thereby supporting ethical and governance-oriented approaches to sustainable AI development.

Keywords: Sustainable AI, Responsible AI, Carbon Footprint of AI, Emissions Transparency, Energy-Aware Machine Learning

1. Introduction

Artificial Intelligence (AI) has experienced rapid expansion and has emerged as a cornerstone of daily life, facilitating ease of use and automation across a wide variety of applications. However, this advancement incurs significant environmental costs. Recent studies indicate that training large-scale deep learning models can produce CO_2 emissions on a scale comparable to major industrial or transportation

activities (Bolón-Canedo et al., 2024). The training of machine learning (ML) and deep learning (DL) models necessitates intensive computation in short bursts utilizing Graphics Processing Units (GPUs) or Tensor Processing Units (TPUs). The deployment of models at scale further amplifies the energy demand throughout their operational lifetime (Bolón-Canedo et al., 2024). Consequently, both training and deployment contribute substantially to energy consumption and greenhouse gas (GHG) emissions. Systematic measurement and transparent reporting of these factors are increasingly vital for researchers, developers, and policymakers.

The carbon footprint of computing encompasses the GHG emissions associated with electricity and resources utilized during operation. This footprint is typically calculated by multiplying energy usage in kilowatt hours (kWh) by the carbon intensity of the regional electricity grid, subsequently adjusted for data center overhead, such as cooling, defined by Power Usage Effectiveness (PUE). The credibility of emissions reporting relies on the clear delineation of systems, verifiable emission factors, and the disclosure of uncertainties (Falk et al., 2024). Complications arise from the fact that AI workloads span multiple clouds and accelerators, with emissions varying widely based on the timing and location of computations. As high-performance computing centers expand to accommodate AI workloads, the quantification of emissions becomes increasingly critical. These centers demand additional electricity, placing strain on energy systems. Operators can mitigate emissions through enhanced efficiency in cooling systems, the procurement of clean energy, and the strategic placement of computing centers in decarbonized regions (Silva et al., 2024).

The carbon footprint of AI is predominantly influenced by engineering and infrastructure decisions. For AI practitioners, effective measurement serves as a foundational step. Monitoring energy consumption and emissions within training pipelines renders environmental impacts visible. By instrumenting experiments, recording time, hardware usage, and power consumption, and converting energy into Carbon Dioxide Equivalent (CO_2e) based on regional emission factors, both computational efficiency and accuracy can be evaluated (Martínez et al., 2023). Open-source initiatives such as CodeCarbon and CarbonTracker assist in quantifying energy usage and reporting emissions. However, these tools currently lack a comprehensive dashboard for comparing regions, testing PUE sensitivity, or assessing model-agnostic training bundles. Furthermore, they do not integrate trade-offs related to accuracy or resource consumption into a unified user interface, limiting their utility in supporting experimental scenarios and transparency.

Emissions reporting plays a crucial role in the ethical dimensions of AI, revealing the trade-offs between accuracy, cost, speed, and environmental impact. This transparency fosters green AI practices aimed at enhancing performance per watt rather than optimizing performance indiscriminately. It also enables the establishment of governance frameworks that require the disclosure of environmental impacts through evidence. This work aims to present a viable framework that incorporates emissions quantification,

regional comparisons, and performance-efficiency analyses into everyday experimentation. A proof of concept illustrates how environmental transparency can be operationalized within the context of AI. Although the MNIST workload employed in this study is relatively small, the findings can be generalized to larger models and more complex training scenarios.

While existing tools such as CodeCarbon and CarbonTracker support basic operational emissions estimation, they do not provide integrated regional scenario analysis, PUE sensitivity testing, or accuracy–emissions trade-off visualization within a unified experimental interface. This limits their usefulness for comparative sustainability evaluation and governance-oriented reporting.

2. Literature Review

Due to the significant energy requirement for training and deploying models, the environmental sustainability of AI development has emerged as a critical research agenda. This involves not only the use, but also production cost and sustainability, which are essential to developing green AI systems. As models scale, computational tasks become larger scale, computing is more costly, consumption of electricity increases, and the generation of GHG emissions become higher (Alnafrah, 2025; Bolón-Canedo et al., 2024; and Moravec et al., 2025). Specialized accelerators like GPUs and TPUs exacerbate the scale of these effects as they are part of high-end data centers with high operational overhead, and they run in large data centers (Silva et al., 2024).

2.1 Measuring Carbon Footprints in AI Workloads

Energy measurement forms the basis for estimating emissions from AI workloads. Emissions are then determined as a function of electricity consumption multiplied by the carbon intensity of the regional grid and adjusted with a PUE adjustment for cooling and distribution overhead (Gupta et al., 2022; Silva et al., 2024). Grid composition varies significantly from region to region and there is a heavy variety of emissions of the same workload when the grid is used (Bailie et al., 2023; Hounyo et al., 2024). There is also heterogeneity of system boundaries from work to work, with some reports concentrating on operational emissions only (Scope 2) and others integrating embodied impacts from hardware and supply chain outputs (Falk et al., 2024; Romeiko et al., 2024). Tools such as CodeCarbon and CarbonTracker are used to make determination more consistent through power draw capturing and usage conversion to CO_2e using region-dependent measurements (Gupta et al., 2022; Martínez et al., 2023). Despite methodological advances, existing measurement tools remain fragmented, with limited integration between telemetry capture, infrastructural context, and performance trade-offs.

2.2 Efficiency and Decarbonization Strategies

Research in sustainable AI highlights opportunities to reduce emissions by improving model efficiency. Techniques such as pruning, quantization, and knowledge distillation reduce model size without

major accuracy loss (Bolón-Canedo et al., 2024; Martínez et al., 2023). Hardware utilization also plays a role. Idle accelerators consume energy even without useful work (Gupta et al., 2022; Himeur et al., 2021). In many deployments, inference dominates long-term energy use, making deployment decisions a major determinant of emissions (Gupta et al., 2022; Moravec et al., 2025). Operational strategies such as regional workload migration and carbon-aware scheduling can reduce emissions without changing the model itself by shifting computation to lower-carbon or renewable-intensive periods (Gupta et al., 2022; Hounyo et al., 2024).

Although this research shows us plenty of routes to lower carbon emissions, there aren't enough practical tools available for researchers to test out these efficiency strategies and evaluate the emissions trade-offs in their day-to-day work.

2.3 Lifecycle Assessment in AI Development

Lifecycle Assessment (LCA) provides a comprehensive structure for examining environmental burdens across model development and deployment. LCA distinguishes between operational and embodied emissions and helps standardize comparison across systems (Romeiko et al., 2024). Focusing only on Scope 2 emissions underestimates full impacts, especially as hardware refresh cycles accelerate due to AI demand (Falk et al., 2024). Incorporating LCA into AI evaluation improves comparability and aligns system design with sustainability goals.

Nevertheless, most tools still concentrate on basic operational metrics and fail to capture the broader environmental considerations that matter when running routine training experiments.

2.4 Data Center Infrastructure and Carbon Accountability

Data center design significantly influences emissions outcomes. PUE varies across facilities and advanced cooling technologies can materially reduce energy consumption (Silva et al., 2024). Cloud providers increasingly market low-carbon compute regions, yet developers often deploy workloads without considering environmental impacts (Alnafrah, 2025; Gupta et al., 2022). Electricity procurement strategies such as renewable energy credits or power purchase agreements can reduce emissions, though their effectiveness differs (Gupta et al., 2022; van Wynsberghe, 2021).

Emissions are heavily influenced by how efficient facilities are and where they source their power, but ML practitioners rarely see this information in the tools they use daily.

2.5 Responsible AI Governance and Ethical Imperatives

Environmental impacts of AI represent an ethical responsibility. Responsible AI frameworks identify transparency, accountability, and harm reduction as core principles (European Commission, 2021). Without emissions disclosure, stakeholders cannot evaluate whether AI benefits justify environmental costs. Researchers argue that environmental reporting should become standard practice in publication and peer review (Alnafrah, 2025; Francisco and Linnér, 2023; Frank, 2021; Nasir et al., 2023; Sipola et al., 2023).

Global distribution is also unequal. Climate-vulnerable regions are disproportionately affected by emissions generated in wealthier countries (Falk et al., 2024; van Wynsberghe, 2021). Recent studies apply LCA to full data center infrastructure rather than only operational energy (Lamnatou et al., 2024; Üçtuğ and Ünver, 2023). However, few empirical tools combine real training telemetry with region-sensitive carbon modeling and PUE scenario analysis.

Developers are still missing user-friendly tools that let them track and report environmental impacts within their normal experimental workflows. Overall, existing research lacks a unified, experiment-level platform that connects real training telemetry with regional emissions modeling, infrastructure sensitivity, and performance-efficiency trade-offs.

2.6 Contribution of the Carbon Footprint Calculator

This work addresses these gaps by providing a model-agnostic platform that integrates emissions accounting into the experimentation process. The calculator enables regional comparison, PUE sensitivity testing, hardware utilization analysis, and accuracy-emissions trade-off evaluation within a single interface. It supports responsible AI implementation by connecting performance outcomes to environmental impact and facilitating transparent reporting. The platform aligns with sustainability and governance expectations that environmental metrics be incorporated into experimentation and decision making (European Commission, 2021). It provides a foundation that can be extended to larger models, real-world workloads, and comprehensive LCA frameworks.

2.7 Related tools and difference from existing work

Existing tools such as CodeCarbon and CarbonTracker already support logging energy usage and estimating operational emissions during AI training. They instrument model code, measure power draw, and convert consumption to carbon dioxide equivalent using regional emission factors (Gupta et al., 2022; Martínez et al., 2023). These tools are usually integrated directly into experiments and report results within the development environment. CarbonMindful builds on this line of work by accepting model agnostic telemetry bundles and providing an external dashboard that recalculates emissions under alternative regional and PUE scenarios. It also links emissions to accuracy and hardware utilization and presents the results through an interactive web interface. In this way the platform complements existing trackers and connects emissions accounting to comparative analysis and interpretability that are emphasized in broader sustainability and LCA oriented studies (Falk et al., 2024; Romeiko et al., 2024; Silva et al., 2024).

3. Methodology

This research employs a quantitative experimental approach in which actual training telemetry is gathered from controlled model executions and examined to assess emissions, regional fluctuation, and efficiency compromises.

3.1 System Architecture and Workflow

The CarbonMindful platform operates as a two-phase pipeline that separates data collection from interactive analysis. The first phase runs inside a Jupyter or Google Colab environment and monitors a supervised learning task. The second phase uses a Streamlit interface to process the run package and visualize metrics.

Workflow sequence:

• train a model while CodeCarbon monitors energy and emissions

• generate a run bundle containing telemetry files

• upload the bundle to the Streamlit interface

• explore regional and PUE scenarios and evaluate accuracy–emissions trade-offs

The training task uses a compact fully connected multilayer perceptron (MLP) on the MNIST dataset. The task is intentionally small to emphasize method and tool design rather than absolute emissions scale. Telemetry captured during training includes elapsed time, accuracy, hardware utilization, and emissions recorded by CodeCarbon. Comparable approaches have been recommended to document energy and emissions alongside performance metrics (Gupta et al., 2022; Martínez et al., 2023; Romeiko et al., 2024). The objective of the experiment is not to statistically compare alternative training runs, but to validate the functional workflow, telemetry integrity, and scenario reconstruction capabilities of the platform; therefore, a single controlled run is sufficient for this proof-of-concept.

3.2 Training Setup and Emissions Logging

Training was performed in Google Colab using PyTorch, torchvision, psutil, and optional NVIDIA Management Library (NVML) support. A fixed random seed improved reproducibility. The configuration specified a project name, run identifier, country for offline tracking, PUE value, sampling frequency, and output paths. The MNIST dataset contains 60,000 training images and 10,000 test images of handwritten digits 0 to 9. Each image is a 28 by 28 grayscale bitmap. Images were normalized and loaded in mini batches using PyTorch data loaders.

The model was a compact fully connected MLP trained with the Adam optimizer and cross entropy loss. Each epoch reported training loss, validation loss, training accuracy, and validation accuracy.

A CarbonMeter context manager wrapped the training loop. Inside this context, CodeCarbon recorded energy and emissions and produced an emissions.csv file with cumulative energy in kWh and CO_2e. A parallel UsageSampler thread collected per second CPU and GPU utilization with psutil and NVML and stored the results in samples.csv. Reported energy and emissions values are subject to measurement uncertainty arising from software-based energy estimation, sampling frequency limits, and hardware reporting variability.

At the end of training, CarbonMeter distributed total energy and total CO_2e across epochs based on each epoch share of total training time and stored these values in epochs.csv. It also generated summary.json containing run metadata including total duration, energy, emissions, PUE, region, and hardware characteristics. This structure follows guidance recommending transparent documentation of assumptions and system context to support interpretability and reuse (Falk et al., 2024; Romeiko et al., 2024).

3.3 Run Bundle Structure

The run bundle contains:

• summary.json: total energy (kWh), total emissions (kg CO_2e), duration, PUE, region, cost fields, and hardware metadata

• epochs.csv: accuracy, loss, duration, and per-epoch energy and emissions

• samples.csv: time series CPU and GPU utilization

The Streamlit interface loads the bundle, verifies required fields, and reconstructs emissions. Total energy is treated as a physical measurement. Emissions may be recalculated based on region or PUE selections without rerunning the model. When epochs.csv is present, the dashboard computes a carbon-efficiency metric in kg CO_2e per accuracy percentage point.

3.4 Regional Comparison Design Rationale

Regional comparison applies the measured energy to different grid carbon intensities and PUE assumptions. Four countries were selected to represent a wide range of electricity system profiles. Canada represents a lower-carbon grid with high renewable supply (Bailie et al., 2023). Mexico represents a medium-intensity system with a mixed fuel portfolio (García Clavel et al., 2024). South Korea reflects a high-intensity grid undergoing decarbonization planning (Ahn and Jeon, 2019; Choo et al., 2024). Mongolia represents a high-intensity extreme dominated by coal (Gansukh, 2021; Lei et al., 2022). This range enables analysis of how identical AI workloads produce different emissions solely due to infrastructure and geography. Regional emissions are calculated using representative grid factors expressed in kg CO_2e per kWh.

Emissions follow the following formula:

emissions = energy × PUE × grid factor

3.5 Visualization and Interpretation Framework

The dashboard includes reference footprint metrics, regional comparison results, PUE sensitivity charts, carbon-efficiency values, accuracy curves, and resource utilization graphs. Carbon efficiency is defined as total kilograms of CO_2e divided by final model accuracy expressed as a percentage point, yielding kg CO_2e per accuracy point. Metric cards summarize total energy, emissions, and PUE. Charts show emissions by region, PUE variation, accuracy by epoch, and utilization over time. The interface is designed to support interpretability and emissions-aware experimentation.

3.6 Ethical Framing and Limitations

From an ethical standpoint, the approach is founded on the principle that environmental consequences represent a dimension of harm that should be rendered transparent and subject to scrutiny within responsible AI governance structures (European Commission, 2021; Falk et al., 2024; van Wynsberghe, 2021). By incorporating emissions monitoring into the training process and by offering a systematic method to evaluate regional and infrastructure alternatives, the system seeks to promote transparency, accountability, and proportionality. The calculator does not determine whether a particular footprint is justifiable but supplies evidence that can be balanced against the advantages of implementing a model in a specific context.

The regional component also relates to equity considerations. Emissions from computing facilities situated in affluent nations contribute to climate consequences that disproportionately impact climate-vulnerable areas, many of which have contributed comparatively little to historical emissions (Falk et al., 2024; Gansukh, 2021). By explicitly demonstrating how the identical AI workload would generate varying amounts of CO_2e in different nations, the tool promotes contemplation about the worldwide distribution of environmental costs associated with AI and the influence of infrastructure choices in reducing or intensifying these costs.

Multiple limitations warrant recognition. First, the emission factors employed for the four nations are static and simplified; they do not represent temporal fluctuation in grid carbon intensity or distinctions between utilities or cloud service providers. Second, the PUE values are user-determined and handled as uniform constants, whereas actual computing facilities may exhibit more intricate and variable energy dynamics (Gupta et al., 2022; Silva et al., 2024). Third, the present implementation concentrates on operational emissions linked to electricity usage and does not measure embodied emissions from hardware production or upstream supply networks, despite literature suggesting that these can be substantial and should be incorporated in thorough lifecycle evaluations (Falk et al., 2024; Romeiko et al., 2024).

Lastly, the demonstration task is founded on a relatively modest image classification challenge. Although the methodological approach is transferable to larger models and practical workloads, absolute footprint scales and compromises would vary in operational environments. The calculator should consequently be regarded as a methodological prototype that demonstrates how emissions monitoring, regional analysis, and basic efficiency indicators can be incorporated into AI development, rather than as a conclusive accounting instrument for any industrial application.

4. Results

This section showcases the results produced by CarbonMindful using real execution data acquired from the construction of a MLP model using the MNIST dataset. Although MNIST is a small task with a short cycle, the experiment has shown the capability and analysis of the platform, which was designed for the

development of any AI or ML model. Because the experiment is based on a single controlled training execution, the results are descriptive rather than inferential and do not provide statistical significance testing. Uploaded training packages, comprising summary.json, epochs.csv, and samples.csv, facilitated the reconstruction of energy usage, emissions calculation, PUE sensitivity examination, hardware resource evaluation, and carbon performance computations.

4.1 Regional Emissions Comparison

Utilizing the identical training run reference energy usage profile of 0.000625 kilowatt hours and a data center PUE of 1.20, the platform computed the corresponding emissions across four national power grids: Canada, South Korea, Mexico, and Mongolia. Based on Figure 1, Canada emitted the least amount of CO_2e with approximately 0.00010 kg, South Korea and Mexico emitted more with approximately 0.00033 kg and 0.00032 kg, respectively. Mongolia had the largest carbon emission with approximately 0.00056 kg. These differences represent the effects of different levels of national power grid carbon intensity, which is influenced by the ratio of the renewable electricity generated compared to fossil fuels. Recent studies examining regional decarbonization trends have verified this heterogeneity and shown that structural differences in energy systems lead to significant difference in corresponding emissions outcomes (Bailie et al., 2023; Choo et al., 2024; García Clavel et al., 2024; Gansukh, 2021).

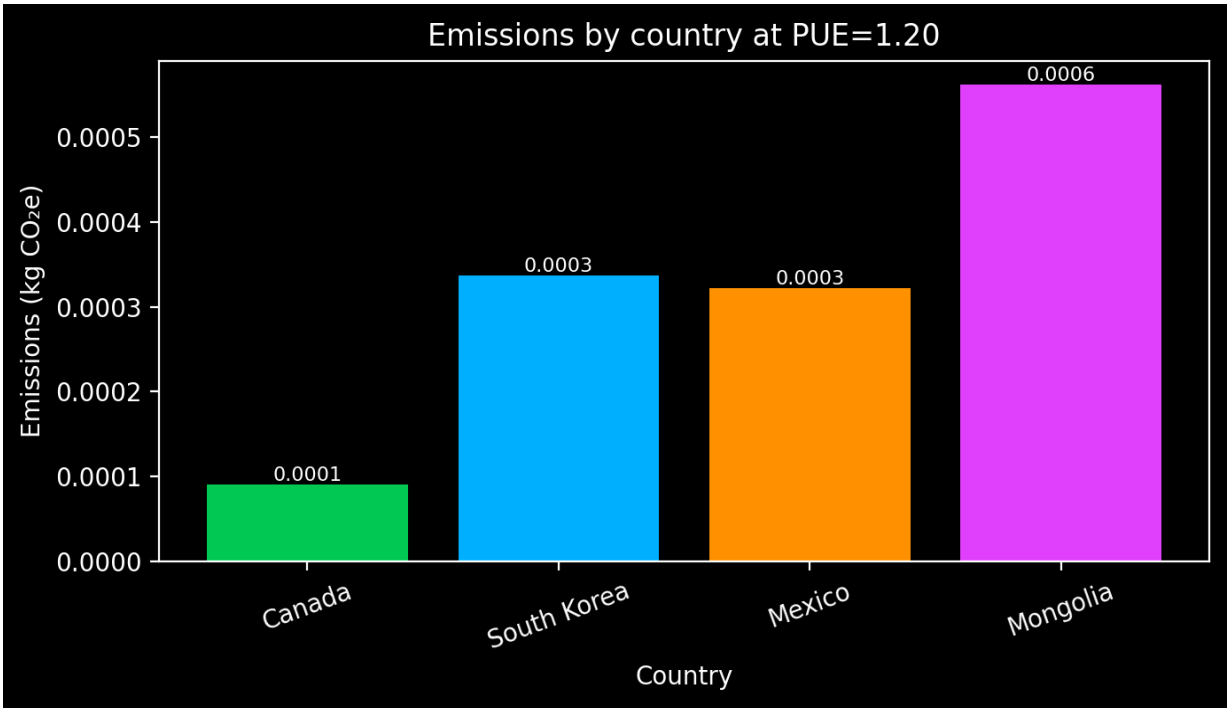


Figure 1. Emissions by country at PUE 1.20

These outcomes reveal that the environmental consequences of AI training cannot be assessed independently from geographical and infrastructural considerations. As highlighted in current sustainable

AI research, regional consciousness and emissions attribution are essential to facilitate transparent and accountable AI implementation (Bolón-Canedo et al., 2024; Falk et al., 2024; Silva et al., 2024).

4.2 PUE Sensitivity

To assess infrastructure dependency, emissions were recalculated employing a less optimal PUE value of 1.60. Figure 2 demonstrates that emissions increased proportionally throughout all four regions while maintaining the identical ranking. For instance, emissions in Mongolia escalated from roughly 0.00056 to 0.00073 kilograms of CO_2e, whereas in Canada the escalation was from roughly 0.00010 to 0.00013 kilograms. The platform also determined the percentage variation, revealing that performing the training run in Canada rather than Mongolia could decrease emissions by roughly 84 percent. These outcomes strengthen findings in recent literature that enhancements in energy efficiency within high performance computing settings can substantially reduce environmental consequences (Silva et al., 2024; Gupta et al., 2022).

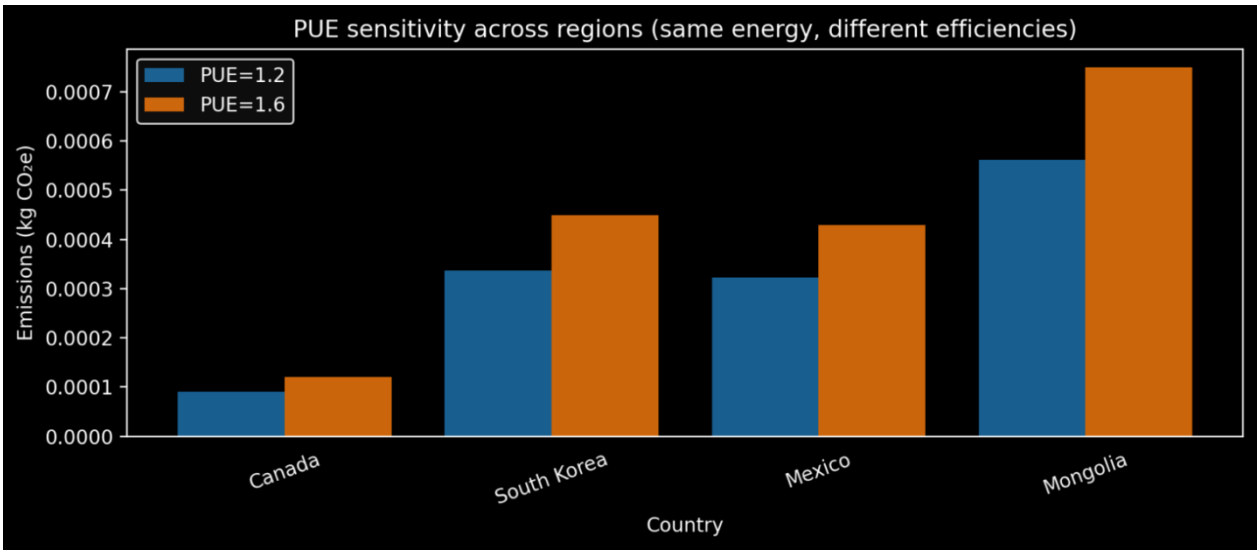


Figure 2. PUE sensitivity comparison at PUE 1.20 and PUE 1.60 across regions

4.3 Carbon Efficiency Metric

The platform determined a carbon efficiency indicator defined as kilograms of CO_2e released per one percent model accuracy. The final model accuracy achieved 95.14 percent, whereas the total emissions for the training run were 0.000269 kilograms of CO_2e, yielding a carbon efficiency value of 0.000003 kilograms per accuracy point. Figure 3 depicts that accuracy advanced from roughly 92.6 percent to 95.1 percent between epoch 1 and epoch 3, whereas emissions increased linearly with duration based on cumulative energy usage. The outcomes reveal diminishing returns, where supplementary accuracy improvements demand disproportionately greater emissions, validating existing guidance that efficiency-conscious training such as adaptive early stopping may be a feasible mitigation strategy (Martínez et al., 2023; Romeiko et al., 2024).

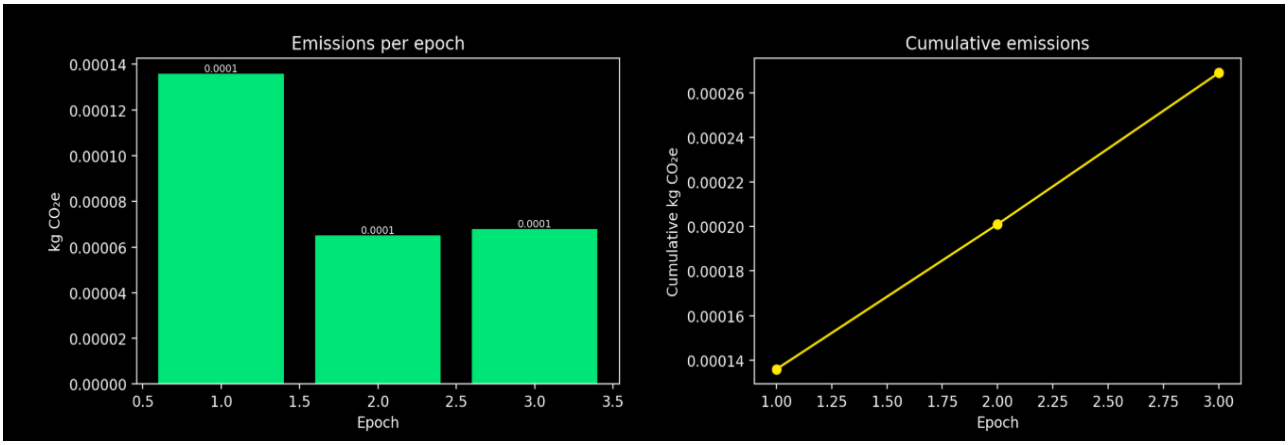


Figure 3. Accuracy by epoch and cumulative emissions

4.4 Hardware Utilization

Hardware resource analysis revealed that CPU usage climbed from roughly 10 percent to maximum capacity and remained near 100 percent for virtually the complete training duration, as displayed in Figure 4. This signifies that the chosen hardware was effectively employed for the workload. GPU acceleration was not required for this workload. However, conclusions regarding GPU efficiency cannot be generalized beyond this lightweight benchmark. The incorporation of utilization diagnostics permits informed resource distribution, promoting responsible alignment of computational resources to task complexity, aligned with sustainability frameworks in AI research (Bolón-Canedo et al., 2024; Falk et al., 2024).

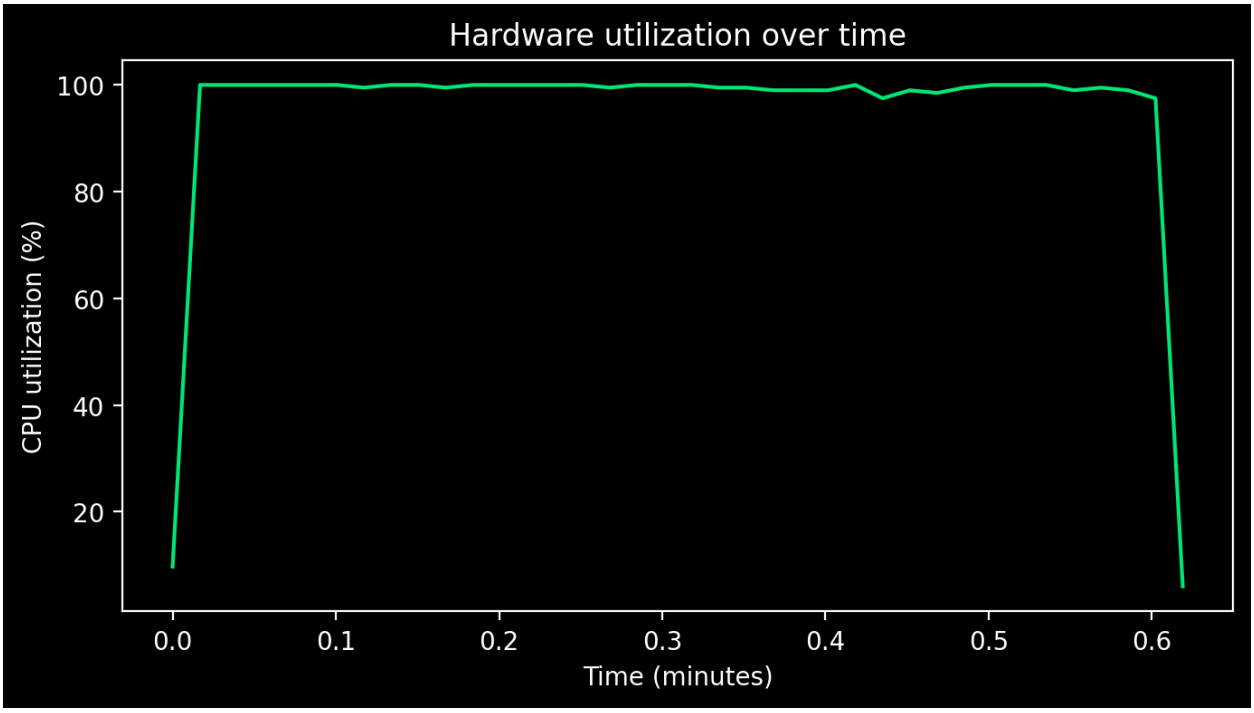


Figure 4. CPU utilization during training

4.5 Summary of Findings

Overall, the outcomes demonstrate that the environmental consequences of model training are determined by an interaction of regional electricity production characteristics, data center infrastructure performance, accuracy-emissions compromises, and hardware resource patterns. The regional analysis reveals that the identical training workload can generate considerably different emissions depending on where it is performed, with Canada producing the lowest footprint and Mongolia the highest under the assessed parameters. The PUE sensitivity examination further illustrates that even minor variations in data center performance amplify or reduce emissions throughout all regions in a predominantly proportional fashion, emphasizing the significance of infrastructure-level choices. At the model level, the carbon efficiency indicator shows that subsequent accuracy improvements require a higher marginal emissions expense, indicating that beyond a particular threshold additional training offers diminishing returns from both a performance and sustainability standpoint. Finally, the hardware resource examination reveals that matching task complexity with suitable resources can avoid unnecessary overprovisioning, since the MNIST workload already maximized CPU capacity and would not clearly gain from more carbon-intensive hardware. Collectively, these outcomes demonstrate that CarbonMindful can deliver actionable understanding across multiple dimensions of the AI lifecycle, allowing practitioners to make more informed and responsible choices about where, how, and on what infrastructure models are trained.

5. Discussion

The empirical outcomes produced through CarbonMindful establish that environmental results linked to ML training are significantly influenced by geographical electricity procurement, data center infrastructure performance and performance compromises at the model scale. Although the demonstration experiment employed a relatively modest MLP developed on the MNIST dataset, the discoveries suggest that even lightweight workloads can display considerable variation in emissions depending on contextual elements surrounding computational infrastructure rather than exclusively on model architecture or dataset complexity.

5.1 Interpretation of empirical findings

The regional comparison element of the platform disclosed that emissions fluctuated substantially across the four investigated national contexts, with Canada generating the lowest emissions and Mongolia the highest. These distinctions align directly to variability in grid carbon intensity recorded across national and regional energy systems (Bailie et al., 2023; Choo et al., 2024; García Clavel et al., 2024; Gansukh, 2021; Hounyo et al., 2024; Lei et al., 2022). The outcomes strengthen findings in energy transition scholarship that highlight the structural heterogeneity of electricity generation portfolios and their consequences for decarbonization trajectories. Within the framework of AI training, the discoveries empirically confirm assertions that infrastructure positioning represents a significant determinant of environmental consequences rather than a neutral logistical decision (Silva et al., 2024; Gupta et al., 2022).

Likewise, the PUE sensitivity outcomes demonstrated proportional escalations in emissions when PUE values shifted from 1.20 to 1.60, revealing that facility-level efficiency performs a considerable role in determining environmental results. Studies at the intersection of HPC and sustainability have underscored the significance of optimizing data center efficiency and power management approaches in reducing aggregate emissions (Silva et al., 2024; Gupta et al., 2022). The calculator's outputs operationalize these discoveries by illustrating measurable consequences through a tangible, experiment-level comparison rather than abstract projection.

At the model-performance scale, the carbon efficiency indicator showed diminishing returns where incremental accuracy enhancements were linked with disproportionately elevated emissions. This observation reflects findings in environmental evaluation research on ML, which warns that escalating computational requirements for marginal performance improvements drives unsustainable energy usage (Martínez et al., 2023; Romeiko et al., 2024). The hardware utilization visualization reinforced this relationship, depicting that CPU resources were nearly completely utilized, suggesting that transitioning to GPU hardware would probably increase carbon expense without providing meaningful performance benefit. These interpretations remain illustrative and should not be directly extrapolated to large-scale industrial AI systems without further empirical validation.

These results collectively indicate that sustainable AI development demands careful coordination between model scale, hardware allocation and performance expectations.

5.2 Alignment with and contribution to sustainable AI research

The findings align with recent research emphasizing the need to integrate environmental perspectives into the design and evaluation of AI systems (Bolón-Canedo et al., 2024; Falk et al., 2024; Silva et al., 2024; van Wynsberghe, 2021). CarbonMindful supports the Green AI objective of evaluating computational efficiency alongside accuracy and provides transparent comparative metrics that help assess accuracy–emissions trade-offs. By visualizing emissions across regions and PUE configurations, the tool offers empirical evidence that supports more environmentally accountable experiment practices.

The outcomes also reinforce arguments that AI operates within broader sociotechnical systems shaped by infrastructure, institutions, and policy rather than purely algorithmic design (Falk et al., 2024; Romeiko et al., 2024). Differences in emissions across countries demonstrate that decisions about where computation occurs carry ethical and geopolitical implications. This connects AI development to global equity discussions, recognizing that environmental burdens and climate impacts are distributed unevenly across regions and populations (Bailie et al., 2023; Lei et al., 2022; Nasir et al., 2023).

5.3 Practical implications for AI development and governance

The findings have practical implications for organizations developing or deploying ML systems. They support recommendations for carbon-aware scheduling and workload placement, showing that substantial emissions reductions can be achieved by selecting low-carbon regions or more efficient data centers without changing model architecture (Gupta et al., 2022; Silva et al., 2024). The carbon-efficiency indicator demonstrates the value of incorporating emissions evaluation into routine experimentation, reinforcing expectations that responsible AI should consider resource use alongside performance metrics (European Commission, 2021; Francisco and Linnér, 2023; Papagiannidis et al., 2025).

The results also encourage organizations to integrate environmental considerations within model selection and infrastructure decisions. Emissions insights can inform risk assessment, infrastructure procurement, and sustainability planning in ways that align operational decisions with broader governance and societal expectations (Di Vaio et al., 2020; König et al., 2023; Nasir et al., 2023; Rodrigues, 2020). Embedding such practices into standard development workflows represents progress toward institutionalizing sustainable approaches to AI rather than treating them as optional or purely conceptual objectives (Alzoubi and Mishra, 2024).

5.4 Future directions suggested by the results

Although the present implementation establishes feasibility, the outcomes reveal opportunities for methodological advancement. One trajectory is the integration of temporal variability in electricity emissions, permitting training planning based on seasonal or hourly renewable availability. A second

trajectory concerns cloud-specific or provider-level emissions factors, which would enable closer correspondence with real-world deployment scenarios. Additionally, incorporating embodied emissions associated with hardware production and data center construction would address recommendations in sustainability research regarding comprehensive lifecycle assessment (Falk et al., 2024; Romeiko et al., 2024; Moravec et al., 2025; Alnafrah, 2025). Finally, implementing the platform for larger models would supply insight into the degree to which observed behavioral patterns scale beyond small datasets and architectures.

6. Ethical Implications

CarbonMindful represents not merely a technical instrument but also an ethical contribution to how AI systems are conceived, developed and assessed. By revealing the energy consumption and emissions linked to training executions, and by connecting these consequences to regional power grids and infrastructure performance, the platform participates directly with current discussions about sustainable AI, global equity and responsible oversight. This section contemplates those ramifications in consideration of recent scholarship on sustainable artificial intelligence, energy infrastructures and AI governance, with a specific emphasis on open access and peer reviewed literature from Elsevier journals and associated publications.

6.1 Environmental sustainability as a central AI ethics concern

Contemporary reviews stress that environmental sustainability must be regarded as a fundamental design priority in AI rather than a secondary limitation that is addressed only after models are implemented (Bolón-Canedo et al., 2024; Silva et al., 2024). These investigations underscore the swift expansion in computational requirements, and the corresponding rise in electricity usage and GHG emissions, as ML models grow larger, more abundant and more extensively implemented. Simultaneously, an expanding collection of scholarships in energy infrastructures and environmental oversight indicates that the environmental consequences of digital technologies are frequently underreported or concealed by aggregate indicators that do not represent the contribution of individual systems or workloads (Alnafrah, 2025; Francisco & Linnér, 2023; Moravec et al., 2025).

The calculator addresses directly this issue by operationalizing environmental sustainability at the scale of specific training executions. Rather than treating sustainability as an abstract concept, the platform calculates emissions in kilograms of CO_2e, disaggregated by region and PUE scenario, and connects these figures to model effectiveness through a carbon efficiency indicator. This corresponds with appeals for quantifiable criteria and metrics for sustainable AI that regard AI as a sociotechnical system integrated in energy infrastructures and institutional frameworks (Alnafrah, 2025; Romeiko et al., 2024). In this regard, the tool facilitates the normative transformation characterized in sustainable AI literature, where the

objective is not merely to utilize AI for sustainability initiatives but also to guarantee the sustainability of AI itself (Bolón-Canedo et al., 2024; Nasir et al., 2023).

6.2 Transparency, explainability and responsible reporting

Transparency and accountability constitute fundamental principles in AI ethics and governance structures, yet they are frequently articulated in broad terms without clarifying how environmental consequences should be recorded and conveyed (König et al., 2023; Liao et al., 2024; Rodrigues, 2020). Much initial scholarship on AI ethics concentrated on concerns such as bias, privacy and human rights, while environmental impacts remained relatively overlooked. Contemporary examinations of AI governance emphasize the necessity of progressing beyond abstract principles toward tangible practices and instruments that render harms and compromises visible to both technical and non-technical participants (Papagiannidis et al., 2025; Di Vaio et al., 2020).

CarbonMindful advances this initiative by delivering run-level transparency regarding energy consumption and emissions. The outputs generated in the MNIST demonstration encompass total energy in kilowatt hours, total emissions in kilograms of CO_2e, emissions by nation under varying grid parameters, and the responsiveness of these figures to PUE. These indicators can be extracted, recorded and appended to experiment documentation or model cards, thus expanding the boundaries of transparency beyond accuracy, loss and latency toward environmental effectiveness. This form of documentation is aligned with wider guidance that organizations should incorporate environmental data into their AI documentation and disclosure procedures (Alnafrah, 2025; Francisco & Linnér, 2023; Nasir et al., 2023).

The visualizations generated by the platform also facilitate explainability. By displaying emissions for Canada, South Korea, Mexico and Mongolia concurrently, and by demonstrating the consequence of transitioning from PUE 1.20 to PUE 1.60, the tool enables decision makers to comprehend how geographical and infrastructural elements shape the environmental ramifications of training. This form of interpretability is essential for governance mechanisms that encompass a spectrum of participants, including management, policy makers, regulators and, in certain instances, affected populations (König et al., 2023; Papagiannidis et al., 2025). The calculator consequently operates as a connection between technical experimentation and governance-focused transparency.

6.3 Justice and the global distribution of carbon burdens

A significant ethical consequence of region-aware emissions modeling pertains to concerns of justice and the worldwide distribution of carbon burdens. Investigations of national and subnational energy transitions reveal that electricity grid carbon intensity fluctuates considerably between countries and regions, contingent on their energy compositions, policy determinations and economic frameworks (Bailie et al., 2023; Choo et al., 2024; García Clavel et al., 2024; Gansukh, 2021; Hounyo et al., 2024; Lei et al., 2022). The outcomes of the MNIST case study mirror this variability: when the identical training workload is

simulated on different grids, the emissions in Mongolia are multiple times greater than those in Canada, with South Korea and Mexico in intermediate standings.

From an ethical standpoint, this introduces the question of who shoulders the environmental ramifications of AI development. If organizations elect to situate compute-intensive workloads in regions with elevated grid carbon intensity due to cost, regulatory or logistical benefits, the resultant emissions contribute to local and global climate hazards that disproportionately impact some populations more than others. Environmental justice scholarship has extensively observed that marginalized communities frequently encounter higher exposure to environmental damages, while obtaining fewer of the corresponding advantages (Bailie et al., 2023; Lei et al., 2022). In the realm of AI, a comparable pattern could materialize if carbon-intensive computing is concentrated in countries or regions while the economic gains of AI accumulate elsewhere.

The calculator does not endeavor to resolve these structural inequities, but it does render them more apparent. By measuring how emissions transform when the identical training run is hypothetically performed in different countries, and by calculating percentage distinctions such as the roughly 84 percent reduction when transitioning from Mongolia to Canada in the demonstration scenario, the tool encourages users to contemplate the ethical geography of their computing choices. These contemplations are compatible with wider appeals to harmonize AI development with sustainable development goals, including goals associated with climate action, reduced inequalities and responsible consumption and production (Di Vaio et al., 2020; Nasir et al., 2023). The platform can consequently facilitate more informed consideration about where large-scale AI development should be positioned and under what circumstances.

6.4 Carbon metrics, trade-offs and their limitations

The incorporation of a carbon efficiency indicator in the platform, characterized as kilograms of CO_2e per one percentage point of model accuracy, establishes a compelling method to visualize compromises between performance and environmental consequences. In the MNIST experiment, the indicator demonstrates that subsequent epochs yield diminishing returns: accuracy advances from roughly 92.6 percent to 95.1 percent across three epochs, while cumulative emissions escalate approximately linearly with duration. This pattern indicates that beyond a particular threshold, supplementary training may produce modest performance improvements at a considerable environmental expense. Comparable trade-off examinations appear in life cycle and environmental evaluations of ML systems, which advocate for weighing environmental expenses alongside accuracy when assessing model architectures or training approaches (Martínez et al., 2023; Romeiko et al., 2024).

Nevertheless, carbon indicators also possess ethical constraints that must be recognized. First, they represent only one aspect of environmental consequence, concentrating on operational emissions from electricity consumption during training. They do not directly incorporate embodied emissions from

hardware production, infrastructure development, or the broader supply chains participating in producing and cooling DCs and networks (Alnafrah, 2025; Moravec et al., 2025). Second, the acceptability of accuracy-emissions compromises is substantially context dependent. In safety-critical domains such as medical diagnosis or autonomous driving, a minor accuracy reduction may be ethically unacceptable even if it produces a meaningful decrease in emissions. Conversely, in lower-stakes applications, moderate reductions in accuracy may be warranted to prevent unnecessary carbon generation.

The ethical application of carbon indicators consequently demands careful interpretation. Sustainable AI frameworks highlight that indicators should guide decision making but not substitute human judgment or wider normative evaluation (Alnafrah, 2025; Papagiannidis et al., 2025; Silva et al., 2024). The calculator facilitates this position by offering transparent numbers and visualizations without dictating specific thresholds or optimization approaches. Users can implement early stopping, architecture or hyperparameter modifications in manners that honor the contextual requirements of their application while still accounting for environmental consequences. In doing so, the tool advances the cultivation of more sophisticated, multi-dimensional evaluation practices in AI.

6.5 Governance, organizational responsibility and institutionalization

The ethical ramifications of the platform also involve questions of organizational accountability and governance frameworks. Analyses of AI governance reveal that numerous organizations articulate commitment to ethical principles but encounter difficulty transforming these into tangible practices integrated in technical workflows, decision mechanisms and accountability structures (Di Vaio et al., 2020; Papagiannidis et al., 2025). Concurrently, examinations of policy and governance discussions surrounding AI and sustainable development demonstrate that institutions such as the European Union, the United Nations and the World Economic Forum progressively connect AI to wider sustainability initiatives but often lack concrete instruments to track advancement or evaluate compromises (Francisco & Linnér, 2023).

CarbonMindful can function as one component in such governance systems. At the organizational scale, it delivers a mechanism to institutionalize environmental documentation for AI projects by making it straightforward to monitor energy, emissions, regional parameters and PUE configurations for each experiment. Organizations could encourage teams to document every model they're considering for deployment through the platform, tracking its emissions alongside standard performance metrics. These data could subsequently contribute to sustainability reports, internal dashboards or external evaluations, facilitating more credible assertions about the environmental effectiveness of AI portfolios (Alzoubi & Mishra, 2024; Papagiannidis et al., 2025).

From a governance standpoint, such procedures can help narrow the divide between citizens' preferences for more stringent regulation of AI and the current restricted regulatory supervision of environmental consequences (König et al., 2023; Rodrigues, 2020). If organizations embrace transparent

documentation on AI-related emissions and integrate this information into risk management and decision mechanisms, they are better equipped to address emerging regulations and societal expectations. Additionally, linking emissions calculations to sustainable development goals and corporate social responsibility obligations can help synchronize AI initiatives with wider strategic priorities in climate and sustainability (Di Vaio et al., 2020; Nasir et al., 2023).

6.6 Reflection on the role of the platform in responsible and sustainable AI

The developed calculator advances responsible and sustainable AI by making environmental impacts measurable at the level of individual training runs and by enabling emissions-aware experimentation. It provides interpretable indicators that connect technical choices to environmental and ethical implications and supports more informed decision making in AI development (Bolón-Canedo et al., 2024; Papagiannidis et al., 2025). The platform highlights that environmental consequences are relevant across the full spectrum of model sizes, since even small workloads demonstrate meaningful variation across regions and data center efficiencies.

The tool also encourages critical engagement with the broader context in which AI operates, including questions about resource allocation and global variability in carbon intensity (Bailie et al., 2023; García Clavel et al., 2024). Its educational role underscores the value of incorporating sustainability considerations into routine experimentation rather than treating them as post hoc evaluations.

Although the tool presents a useful direction, its impact depends on how it is integrated into real workflows. Without authentic adoption, emissions metrics risk becoming symbolic rather than influencing operational choices, an issue identified in sustainability and technology governance research (Moravec et al., 2025). When combined with organizational accountability frameworks, the platform can help guide decisions about model architecture, training duration, and compute location in ways aligned with environmental and social responsibility (Silva et al., 2024).

7. Conclusion

This report presented CarbonMindful, a platform designed to support responsible and sustainable AI by measuring and visualizing the environmental impact of ML training. The system reconstructs emissions from actual execution telemetry and applies regional and PUE scenarios to show that environmental outcomes vary significantly based on infrastructure and geography. The demonstration using a compact MLP model on the MNIST dataset revealed that identical workloads produce different carbon results depending on the electricity grid and data center efficiency of the selected region.

The PUE sensitivity results illustrated that even modest adjustments in data center performance alter emissions proportionally across all regions. These outcomes indicate that choices about where computation occurs and under what conditions directly influence sustainability outcomes in AI development.

The carbon efficiency metric highlighted diminishing returns. Subsequent improvements in model accuracy required disproportionately higher emissions, suggesting the value of efficiency-oriented training practices. Hardware resource analysis also indicated that for the MNIST task, CPU resources were already maximized and that shifting to GPU hardware would have increased carbon cost with minimal performance benefit. The ethical significance of the platform lies in its ability to convert environmental impact into measurable metrics that can inform decision making in AI development and governance. It supports transparency, accountability, and more evidence-based evaluation of trade-offs that extend beyond accuracy alone. The tool also illustrates that sustainability considerations apply not only to large scale systems but to everyday experimentation. The current version focuses on operational emissions only and does not incorporate embodied emissions associated with hardware production or data center construction, and the MNIST task represents a modest scale. Future work can incorporate cloud specific emission factors, temporal variability in grid emissions, deployment phase energy, and larger models to reflect real world conditions more accurately.

In summary, CarbonMindful demonstrates that environmental sustainability can be integrated into the AI development lifecycle through transparent and quantitative feedback. By exposing environmental consequences at the experimental level, the platform establishes a foundation for more responsible decision making and supports alignment between AI performance objectives and sustainable outcomes.

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